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September 12, 2026

The Editor-in-Chief
Physical Communication, Elsevier

Dear Editor,

I submit for consideration the manuscript “*Evaluated Under What? A Channel-Realism Audit of Semantic Communication for Non-Terrestrial Networks*” as a regular research paper.

Surveys of semantic communication for non-terrestrial networks map each constraint of the satellite link onto the works said to address it. The paper tests that mapping by measuring both of its sides: it propagates public orbital elements to fix the magnitude of each effect from geometry rather than assumption, codes the works a recent survey lists as representative according to whether an effect appears in a paper’s framing or in its evaluation, and applies the measured conditions to a codec trained under the assumption the population most often makes.

Two of its results run against its own premise and are reported as such: the link budget moves slowly enough that a static signal-to-noise ratio within a transmission block is sound, and a codec trained at one operating point loses only about three decibels across a full pass. The cost appears instead on an axis no work in the population instantiates, and what governs it turns out to be the interval over which carrier phase is allowed to accumulate rather than Doppler as such, with the failure threshold scaling as the model predicts across four decades of that interval.

The contribution sits at the physical layer: it is an evaluation of channel assumptions, carrier phase accumulation and residual frequency offset for a learned source-channel code over a low-Earth-orbit link, and this journal has published both semantic satellite communication and experimental analyses of non-terrestrial links. The checklist, the coding sheet carrying the sentence behind every decision, the orbital generator and the evaluation harness are released at <https://github.com/haodpsut/ntn-semcom-realism-audit> with the raw result files, and every number in the manuscript is generated from them. The manuscript states which machine produced which measurement, and reports the rate at which training runs failed a validity criterion declared before the experiments were run rather than reporting only the runs that succeeded.

The manuscript is original, has not been published before, is not under consideration by any other journal or conference, and is single-authored. For transparency I disclose one related manuscript of mine currently under review, a study of quantum federated learning with Kolmogorov-Arnold networks at *Quantum Machine Intelligence*: it shares the audit genre but concerns a different method family, application and population, and shares no data, figure or text with this submission.

I declare no competing interests. No generative model is an author; the Acknowledgment section states where such tools were used.

Yours sincerely,

Phuc Hao Do